

AWS SOC Monitoring Lab

CreateUser IAM Detection Workflow

CloudTrail • CloudWatch • SNS • SQS • Splunk SIEM

Project Overview

This report documents the end-to-end implementation of real-time IAM user creation monitoring in AWS, aligned to SOC detection engineering practices. The workflow detects the CreateUser API event and routes it through an automated alerting and SIEM ingestion pipeline.

IAM user creation is a high-signal event in cloud environments — it may indicate privilege escalation, persistence establishment, or insider threat activity. Detecting it in near real-time is a foundational SOC capability.

Detection Stack

Event Source	AWS CloudTrail — API activity logs
Log Transport	CloudWatch Log Group (aws-cloudtrail-logs-nextwork)
Detection Method	CloudWatch Metric Filter on CreateUser event
Alert Mechanism	CloudWatch Alarm → SNS Topic → Email
SIEM Integration	Splunk Enterprise — index=* sourcetype=aws:cloudtrail
Monitored Event	IAM CreateUser API call
Namespace	SecurityMonitoring
Metric Name	IAMUserCreated

Workflow Summary

- Simulate activity: Create IAM user → triggers CreateUser CloudTrail event
- Verify event in CloudTrail Event History
- Access CloudWatch Log Groups, confirm log stream presence
- Create Metric Filter: { (\$eventName = "CreateUser") }
- Verify IAMUserCreated metric in SecurityMonitoring namespace
- Create CloudWatch Alarm: Sum ≥ 1, Period = 5 min
- Configure SNS topic for email notification
- Deploy alarm, confirm IAM-User-Creation-Alert created
- Validate: SNS email received + Splunk SPL query returns events

Step 1 — Simulate Suspicious Activity: Create IAM User

Create a new IAM user inside AWS IAM. This action generates a CreateUser API event in CloudTrail logs, serving as the initial trigger for the entire detection pipeline.

Objective: *Trigger a real CreateUser CloudTrail event to validate detection coverage.*

Navigation Path

► IAM → Users → Create user

Configuration

User name	Security-engineer (or any test name)
Console access	Enabled (optional for testing)
Password type	Custom password
Permissions	Attach policy directly → AdministratorAccess (for testing)

CreateUser

Create user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Specify user details

User details

User name

Security-engineer

You can enter up to 64 characters. Valid characters: A-Z, a-z, 0-9, and -, ., @, _ (hyphen)

☒ Provide user access to the AWS Management Console - optional

In addition to console access, users with `SigntIn.amazonaws.com:console` permissions can use the same console credentials for programmatically access without the need for access keys

Console password

☐ Autogenerated password

You can view the password after you create the user

☒ Custom password

Enter a custom password for this user

xxxxxxxxxxxx

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols: `!@#$%^&*~`|;:'",-. _{} [] ~!@#$%^&*~`|;:'",-. \_{} [] ~!@#$%^&*~`|;:'",-. _{} []`
- Show password

☒ Users must create a new password at next sign-in - Recommended

Users automatically get the `AWSCognitoChangePassword` policy to allow them to change their own password

 If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.

[Learn more](#)

Create

Next

Home

Users > Users pages > Create user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best practice way to manage user's permissions by job functions. [Learn more >](#)

Permissions options

☐ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a user instead. Then, add the user to the appropriate group.

Permissions policies (1/1493)

Choose one or more policies to attach to your new user

Filter by Type

All types

50 matches

Policy name >	Type	Attached entities
☐ AmazonEC2ContainerRegistryFullAccess	AWS managed	0
☐ AmazonEC2ContainerRegistryPowerTool	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFull	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFullOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFullOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFullOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFullOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFullOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnlyFullOnly	AWS managed	0
☒ AmazonEC2FullAccess	AWS managed	1

Step 1 Specify user details
Step 2 Set permissions
Step 3 **Review and create**
Step 4 Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name Security-engineer	Console password type Custom password	Require password reset Yes
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Permissions summary

Name	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy
IAMUserChangePassword	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Cancel Previous **Create user**

User created successfully
You can view and download the user's password and email instructions for signing in to the AWS Management Console.
[View user](#)

Step 1 Specify user details
Step 2 Set permissions
Step 3 Review and create
Step 4 **Retrieve password**

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

[Email sign-in instructions](#)

Console sign-in URL
<https://922806890303.signin.aws.amazon.com/console>

User name
[Security-engineer](#)

Console password
XXXXXXXXXXXX [Show](#)

Cancel [Download .csv file](#) **Return to users list**

Detection Trigger: Saving the user triggers `AuthorizeSecurityGroupIngress` — AWS records a `CreateUser` event in CloudTrail within seconds.

Step 2 — Verify CloudTrail Event

Confirm that the `CreateUser` event is recorded in AWS CloudTrail Event History. This validates that CloudTrail is correctly capturing IAM API activity before configuring detection rules.

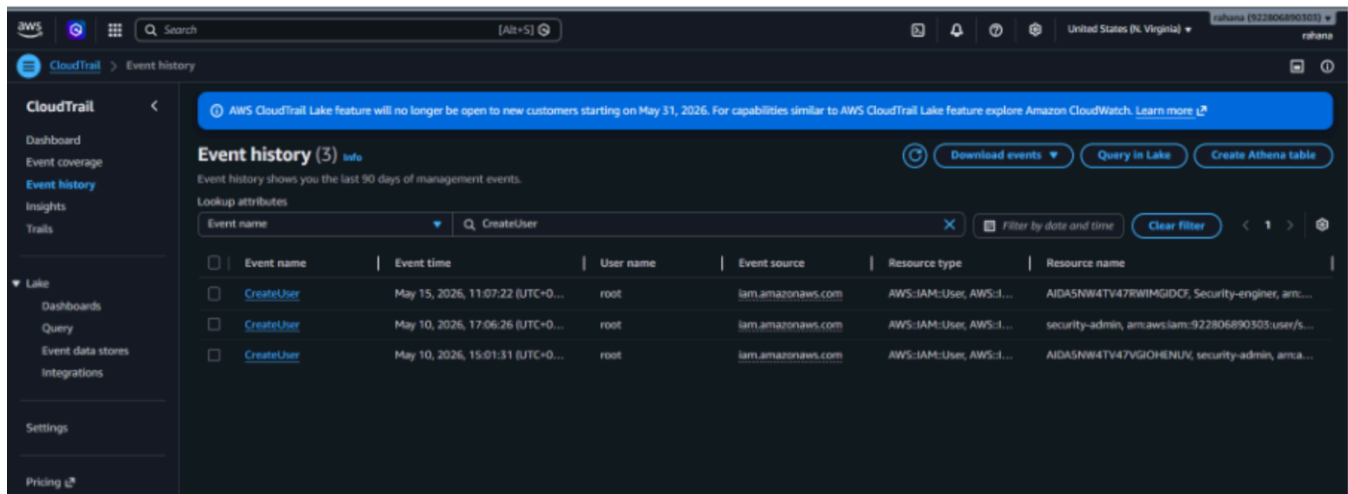
Navigation Path

► [CloudTrail](#) → [Event History](#) → [Search: CreateUser](#)

What to Verify

- Event name: `CreateUser` is present in the event list
- Event time matches when the IAM user was created

- Event source: iam.amazonaws.com
- Resource type: AWS::IAM::User



Note: CloudTrail Event History retains the last 90 days of management events. If no results appear, wait 1–2 minutes for propagation.

Step 3 — Open CloudWatch Log Groups

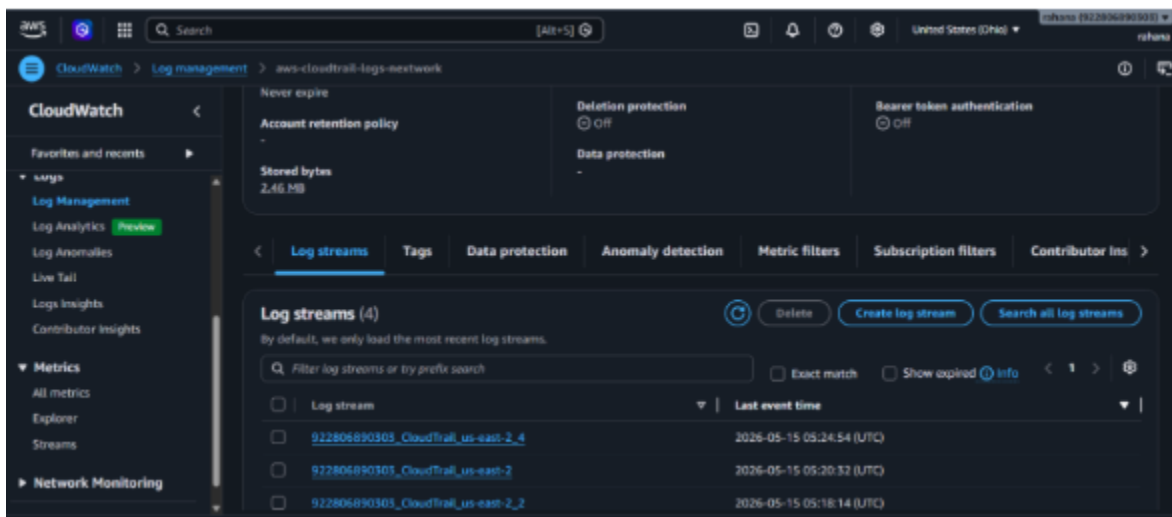
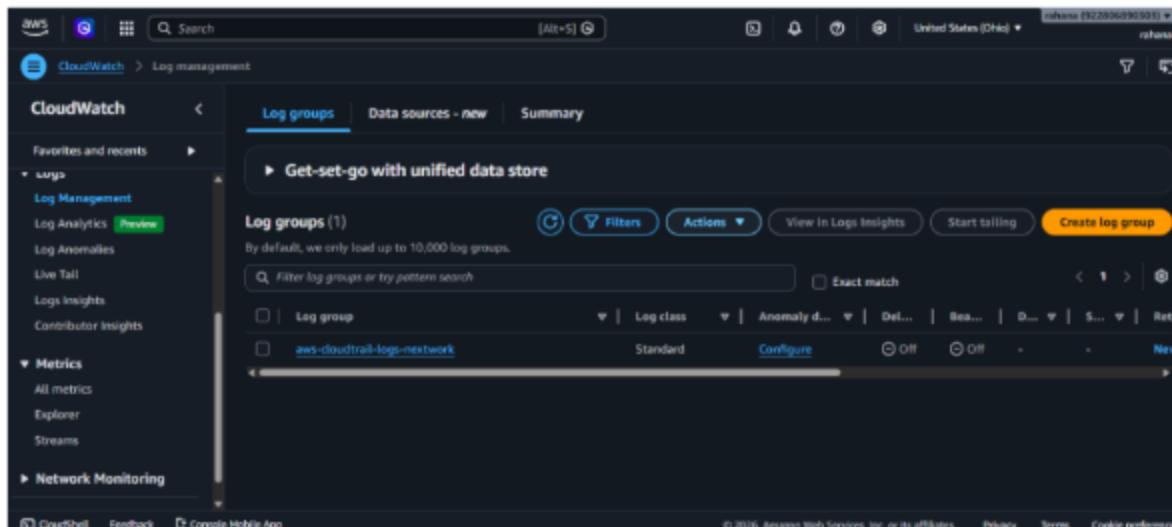
Access the CloudWatch Log Group connected to CloudTrail and verify that CreateUser log events are present in the log streams. This confirms the CloudTrail → CloudWatch integration is active.

Navigation Path

► **CloudWatch** → **Log Management** → **Log groups** → **aws-cloudtrail-logs-nextwork**

What to Verify

- Log group aws-cloudtrail-logs-nextwork is present and active
- Stored bytes > 0 (confirming log data is flowing in)
- Log streams show recent "Last event time" timestamps
- Multiple streams present (one per AWS region/account)



Step 4 — Create CloudWatch Metric Filter

Create a Metric Filter on the CloudTrail log group to parse incoming log events and count occurrences of the CreateUser API call. This converts raw log data into a quantifiable metric.

Navigation Path

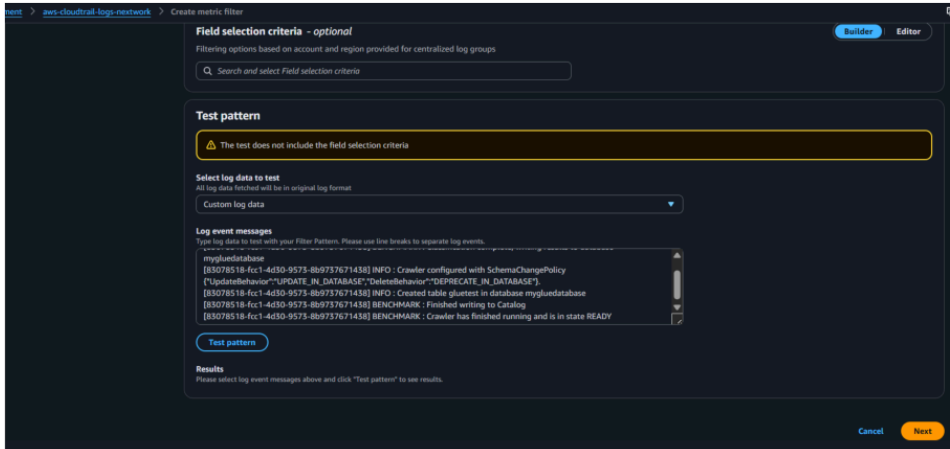
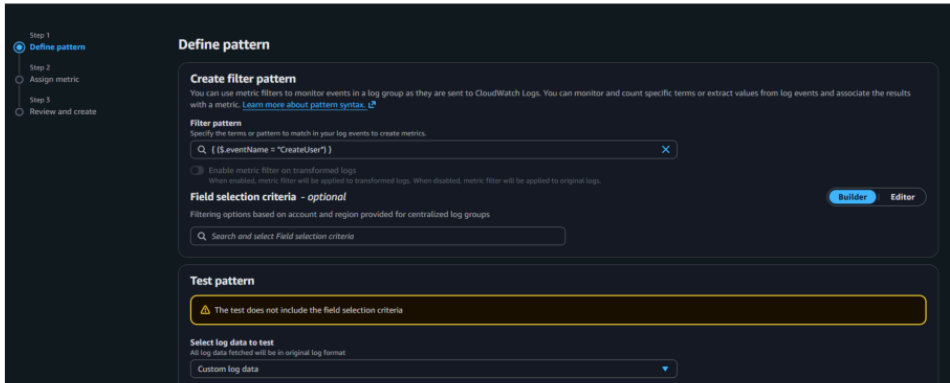
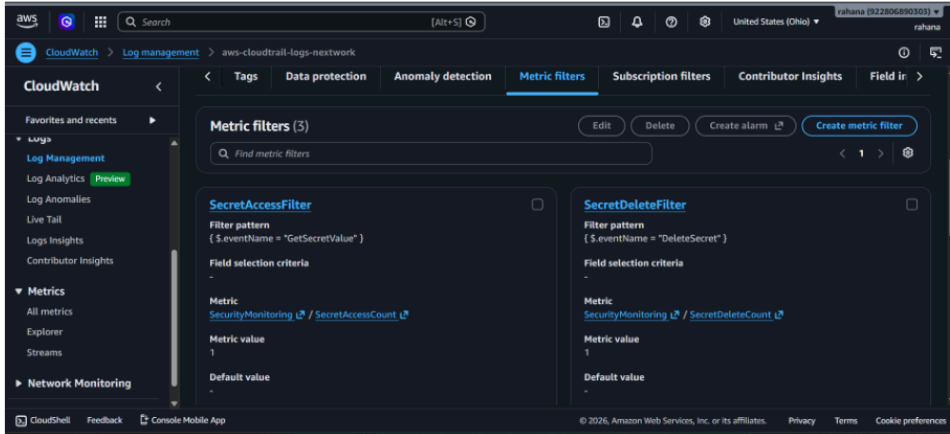
- **CloudWatch** → **Log Groups** → **aws-cloudtrail-logs-nextwork** → **Metric filters** → **Create metric filter**

Filter Pattern

```
{ ($.eventName = "CreateUser") }
```

Metric Configuration

Metric namespace	SecurityMonitoring
Metric name	IAMUserCreated
Metric value	1
Default value	—



CreateUser

4

Best Practice: Always test the filter pattern using the "Test pattern" feature with sample log data before saving, to confirm it matches CreateUser events correctly.

Step 5 — Verify Metric Creation

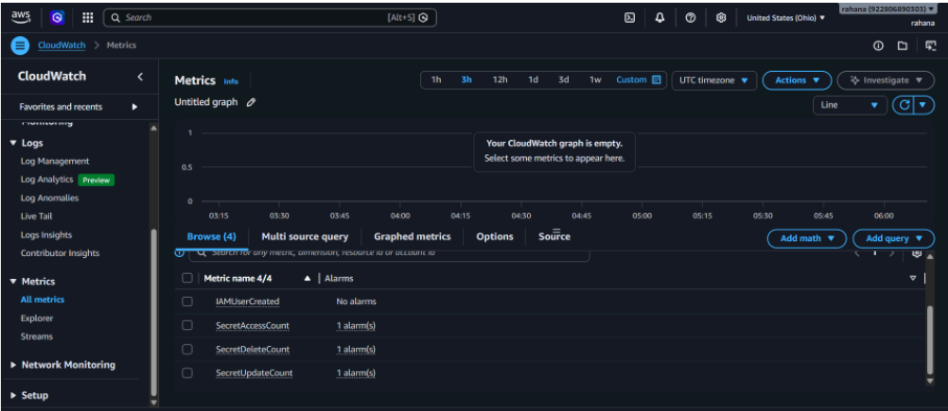
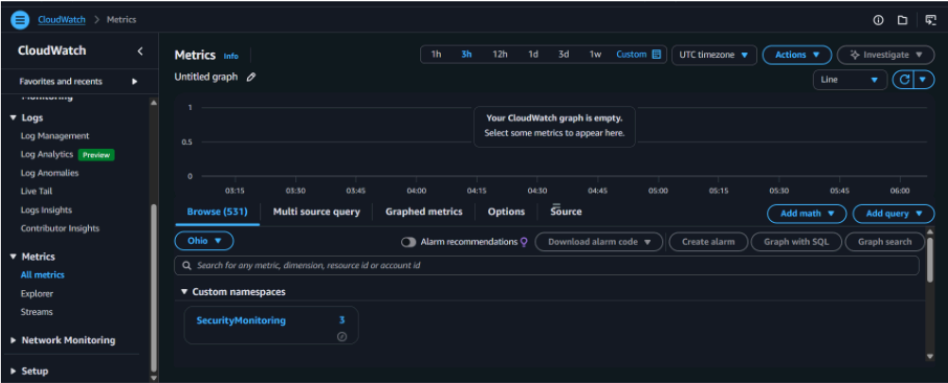
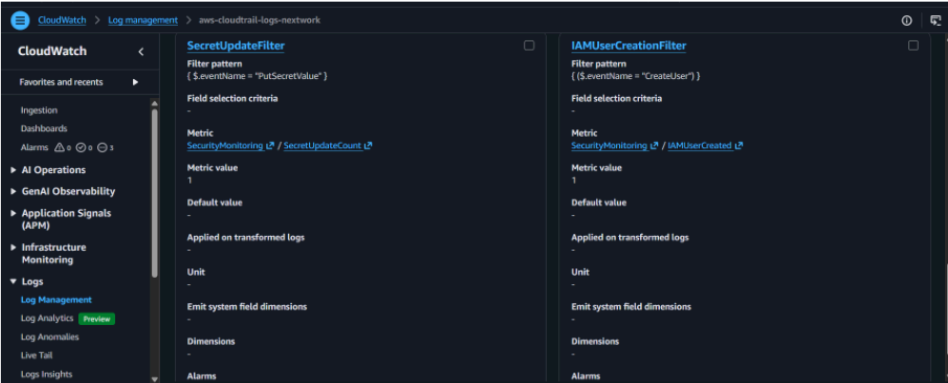
Confirm that the IAMUserCreated metric is successfully created inside the SecurityMonitoring custom namespace and visible in CloudWatch Metrics.

Navigation Path

► [CloudWatch](#) → [Metrics](#) → [All metrics](#) → [Custom namespaces](#) → [SecurityMonitoring](#)

What to Verify

- SecurityMonitoring namespace is listed under Custom namespaces
- IAMUserCreated metric appears alongside other security metrics
- Metric shows "No alarms" initially (alarm is created in the next step)
- Other metrics visible: SecretAccessCount, SecretDeleteCount, SecretUpdateCount



Step 6 — Create CloudWatch Alarm

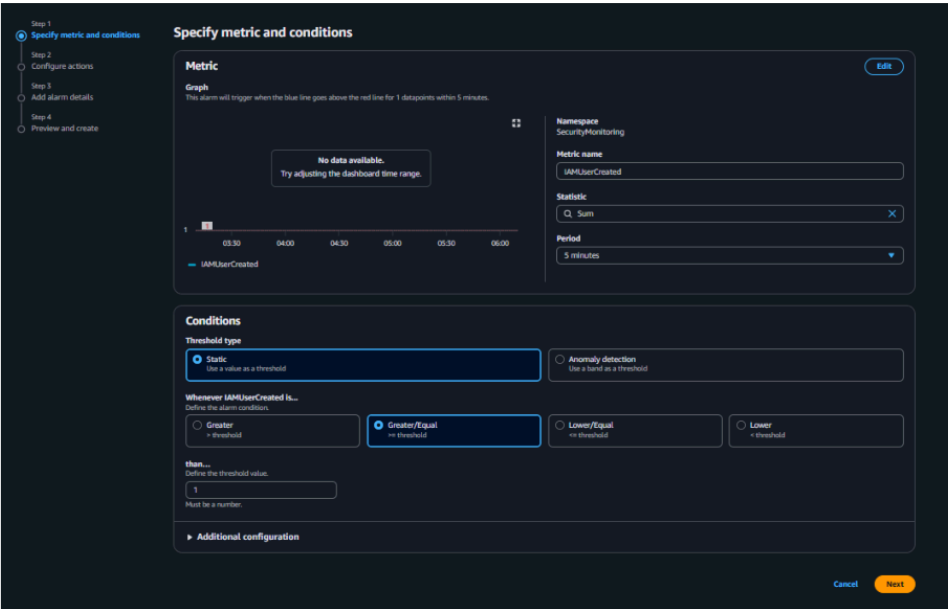
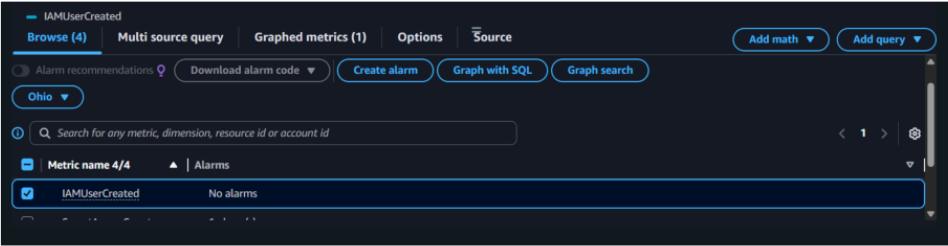
Configure a CloudWatch Alarm on the IAMUserCreated metric to fire whenever a CreateUser event is detected. This is the core alerting trigger in the detection pipeline.

Navigation Path

► [CloudWatch](#) → [Metrics](#) → [SecurityMonitoring](#) → [IAMUserCreated](#) → [Create alarm](#)

Alarm Configuration

Namespace	SecurityMonitoring
Metric name	IAMUserCreated
Statistic	Sum
Period	5 minutes
Threshold type	Static
Condition	Greater/Equal (>=) threshold
Threshold value	1
Alarm name	IAM-User-Creation-Alert



Step 1: Specify metric and conditions
Step 2: **Configure actions**
Step 3: Add alarm details
Step 4: Preview and create

Configure actions

Notification

Alarm state trigger: Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to:

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (required)
email.arn@mycompany.com - View in SNS Console [View](#)

[Add notification](#)

Lambda action

[Add Lambda action](#)

Auto Scaling action

[Add Auto Scaling action](#)

EC2 action

This action is only available for EC2 Per-Instance Metrics.

[Add EC2 action](#)

Systems Manager action [Learn more](#)

This action will create an Incident or Operation in Systems Manager when the alarm is in **alarm** state.

[Add Systems Manager action](#)

Investigation action - NEW [Info](#)

This action will create an investigation when the alarm is in **alarm** state.

[Add investigation action](#)

[Cancel](#) [Previous](#) [Next](#)

Step 1: Specify metric and conditions
Step 2: Configure actions
Step 3: Add alarm details
Step 4: Preview and create

Add alarm details

Name and description

Alarm name:

Alarm description - optional [View formatting guidelines](#)

[Edit](#) [Preview](#)

This is an H1
Double asterisks will produce strong characters
This is [an example](https://example.com/) inline link.

(Up to 1024 characters (0-1024)).

☒ Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

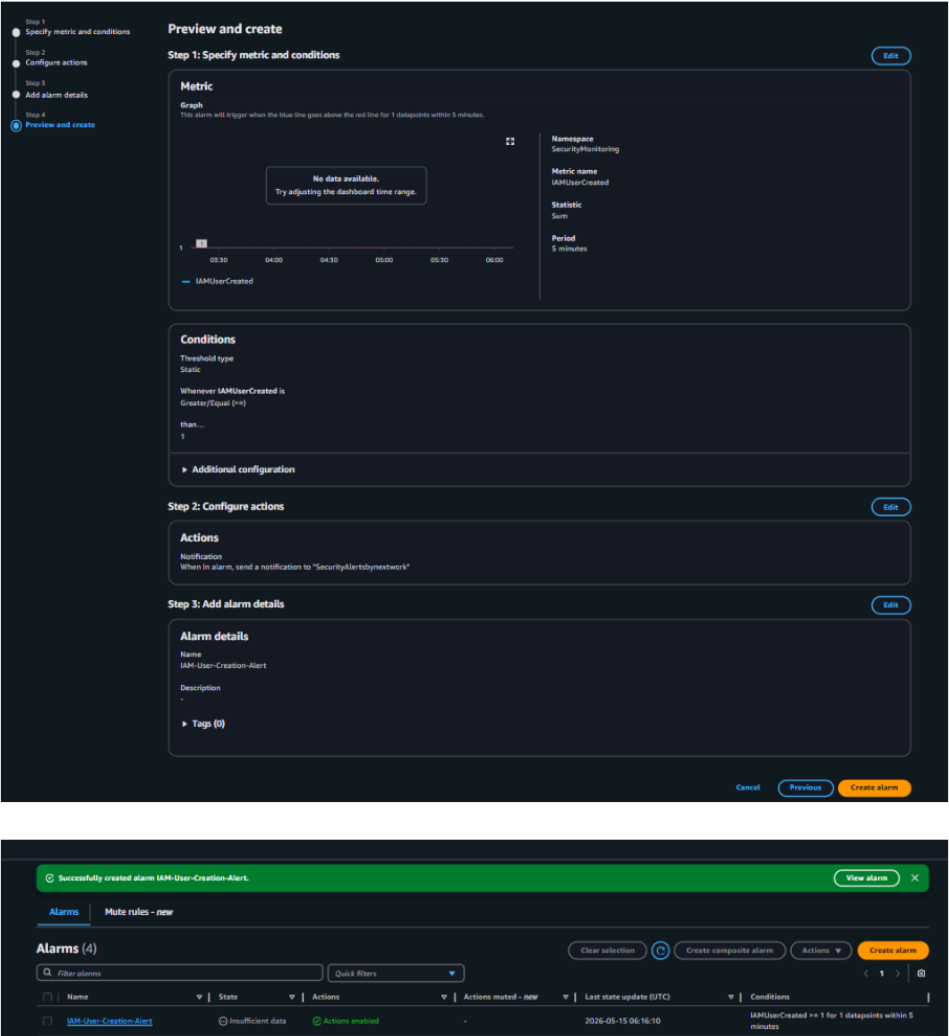
Tags - optional [Info](#)

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 tags.

[Cancel](#) [Previous](#) [Next](#)



Why Sum + Period 5 min? Using Sum ensures even a single event fires the alarm. A 5-minute evaluation window balances detection speed with noise reduction.

Step 7 — Configure SNS Notification

Configure Amazon SNS to send real-time email alerts whenever the CloudWatch alarm enters the ALARM state. This automates analyst notification for every detected CreateUser event.

SNS Configuration

Alarm state trigger	In alarm
SNS topic	SecurityAlertsByNextwork (existing topic)
Notification target	nadhil.ANs@gmail.com
Alarm name	IAM-User-Creation-Alert

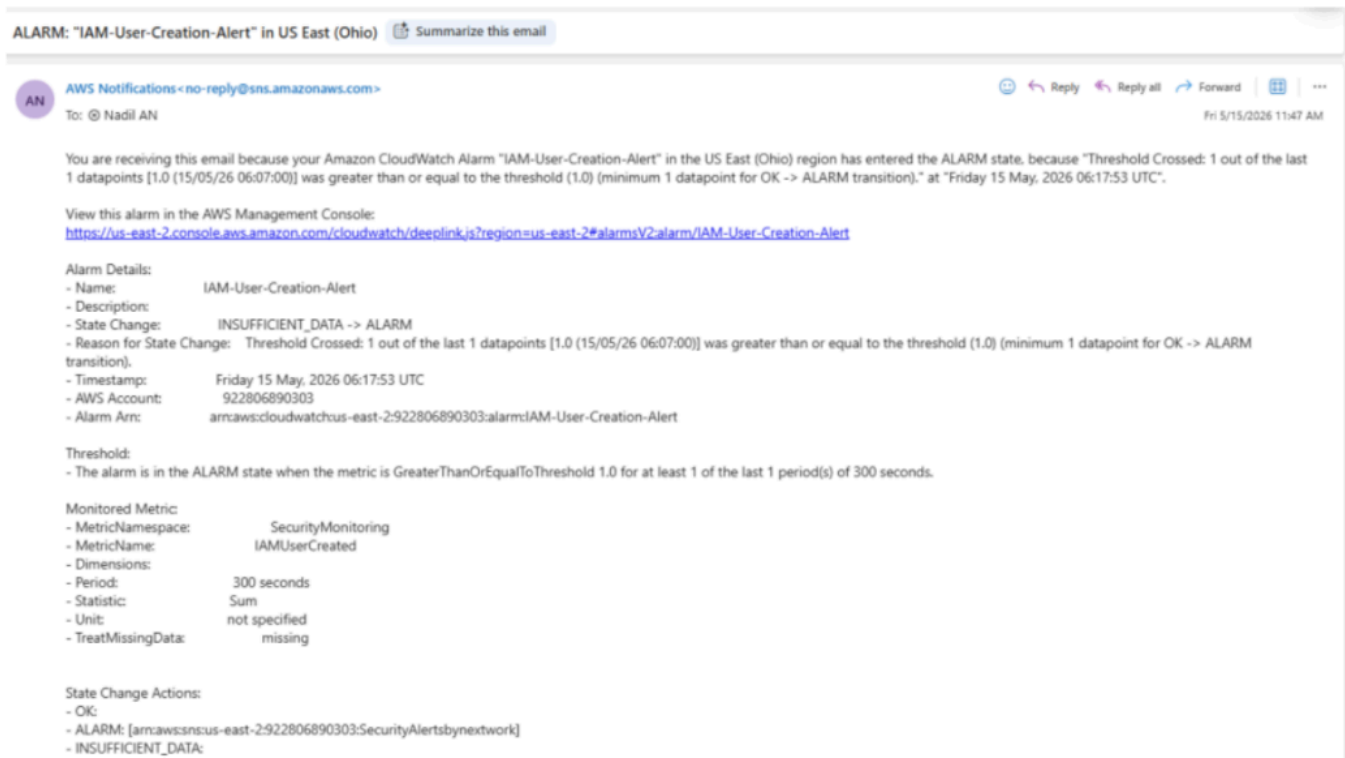
Additional Actions: Lambda actions, Auto Scaling actions, and Systems Manager OpsItem creation can be chained to the same alarm for automated response workflows.

Step 8 — Deploy Detection Alarm

Review the full alarm configuration and deploy. This activates continuous monitoring of IAM CreateUser activity across the AWS environment.

Pre-Deploy Review Checklist

- Step 1 — Metric: SecurityMonitoring / IAMUserCreated / Sum / 5 minutes
- Step 2 — Condition: IAMUserCreated >= 1
- Step 3 — Actions: Send notification to SecurityAlertsByNextwork
- Step 4 — Alarm name: IAM-User-Creation-Alert



Deployment Result: Alarm created successfully. Initial state: Insufficient data (normal — no events yet processed). Actions: Active.

Step 9 — Verify SNS Alert & Splunk Detection

Perform end-to-end validation: confirm the SNS email alert was delivered and verify that the CreateUser event is indexed and searchable in Splunk SIEM.

SNS Email Alert Verification

Check your inbox for the CloudWatch alarm notification email. The email contains full alarm details including threshold values, metric configuration, and timestamps.

Expected Email Details

Subject	ALARM: "IAM-User-Creation-Alert" in US East (Ohio)
State change	INSUFFICIENT_DATA → ALARM
Reason	Threshold Crossed: 1 out of 1 datapoints >= 1.0 (minimum 1 datapoint)
MetricName	IAMUserCreated
Namespace	SecurityMonitoring

Statistic	Sum
Period	300 seconds

Splunk SIEM Validation

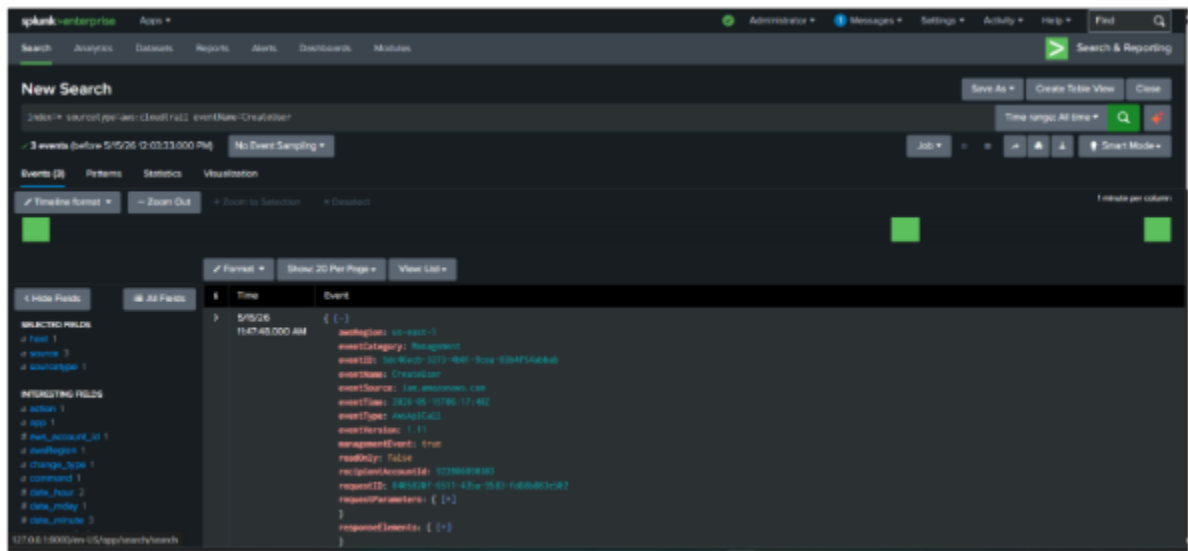
Query Splunk to confirm CloudTrail CreateUser events are being ingested and indexed. This validates the S3 → SQS → Splunk log forwarding pipeline.

SPL Detection Query

```
index=* sourcetype=aws:cloudtrail eventName=CreateUser
```

Expected Results

- Events visible in timeline view with CreateUser events highlighted
- eventName: CreateUser present in all returned events
- eventCategory: Management, eventType: AwsApiCall
- userIdentity, requestParameters, and responseElements populated



Splunk Confirmation: 3 events before 5/15/25 returned by SPL query — confirms CreateUser events are flowing from CloudTrail into Splunk via S3/SQS integration.

Conclusion

A complete AWS SOC monitoring pipeline for IAM user creation activity was successfully implemented and validated end-to-end. The workflow covers event simulation, CloudTrail capture, CloudWatch detection, automated SNS alerting, and Splunk SIEM ingestion.

This implementation demonstrates detection engineering capability aligned to real-world SOC L1 analyst workflows and maps to MITRE ATT&CK Tactic: Persistence (TA0003) — Technique T1136.003: Create Cloud Account.

Outcomes Achieved

- Real-time detection of IAM CreateUser events using CloudWatch Metric Filters
- Automated alerting via CloudWatch Alarms and SNS email notifications
- SIEM integration validated — Splunk ingesting CloudTrail events via S3/SQS
- Full detection pipeline: CloudTrail → CloudWatch → SNS → Splunk confirmed
- Detection rule mapped to MITRE ATT&CK T1136.003 (Create Cloud Account)

MITRE ATT&CK Mapping

Tactic	Persistence (TA0003)
Technique	T1136 — Create Account
Sub-technique	T1136.003 — Cloud Account
Data Source	CloudTrail: CreateUser API call
Detection	CloudWatch Metric Filter: \$.eventName = "CreateUser"